

P.VERB

Dattorro Plate Reverb · Eurorack Module

Stereo in/out · full MIDI · two control pages · 44.1 kHz · ESP32-S3 (AMYboard) · firmware rev-24

Quick start

1. Patch audio into IN (DIG or LINE); take the output from OUT.
2. Set the PAGE toggle to A (teal labels).
3. MIX up, DECAY around the middle. You should hear reverb.
4. The BYPASS toggle kills the effect entirely.

The OLED shows the page you are on and the live value of every control on it.

Front panel

The screen mirrors the panel: the three top knobs appear on the screen's top row, the four bottom knobs on its bottom row. **What you see is where your hand is.**

Control	Type	Function
BYPASS	toggle	Effect in / out. Authoritative — overrides MIDI and serial.
DECAY	slide pot	Reverb time. Page A only; does nothing on Page B.
PAGE A/B	toggle	Selects the control page. Position is the page.
7 rotaries	pots	Dual-function — see the two tables below.

Panel silkscreen: **teal** = Page A, **orange** = Page B.

Page A — the reverb itself

Knob	Label	Range	Notes
slide	DECAY	~0.8 s → freeze	0.8 s at the bottom, ~1.7 s at midpoint, ~3.6 s at three-quarters, infinite hold at full travel.
1	DAMPING	0 – 93 %	High-frequency absorption inside the tank. Higher = darker tail.
2	LOWPASS	0 – 8 kHz	Input low-pass. Fully clockwise = open.
3	HIGHPASS	0 – 1 kHz	Input high-pass. Fully anticlockwise = off.
4	BITRATE	clean → crushed	Bit-depth + sample-rate crush. Fully clockwise = clean. Sits before the tank, so artifacts smear through the reverb.
5	PREDELAY	0 – 240 ms	Gap before the reverb starts.
6	DIFFUSION	0.25 – 0.90	Low = distinct echoes, high = smooth wash.
7	MIX	dry → wet	Wet fraction.

Page B — modulation and levels

The slide pot is unused on this page. Values are restored from memory on power-up, and knobs must be moved past their stored position before they take effect — so a knob left somewhere odd does not snap your settings on page change.

Knob	Label	Range	Notes
1	IN LEVEL	0.2 – 3.0×	Input gain. Not saved — resets each power-up.
2	OUT LEVEL	0.2 – 2.0×	Output gain. Not saved.
3	MOD RATE1	0 – 5 Hz	Tank stage-1 LFO speed.
4	MOD DEPTH1	0 – 32 samples	Stage-1 excursion. Chorusing / detune of the tail.
5	MOD RATE2	0 – 5 Hz	Stage-2 LFO speed, independent of stage 1.
6	MOD DEPTH2	0 – 16 samples	Stage-2 excursion.
7	MIDI CH	1 – 16	RX/TX channel. Saved.

The two LFOs run at deliberately unrelated phases so they never beat together. Small depths thicken the tail; large depths pitch-warble it.

Reading the display

Page A

```
DATTORRO          CPU42%
  DECAY 7.2s
  DAMP LPF  HPF
   65% 4.8k 120
BITR PDLY DIFF  MIX
100% 62ms 71% 35%
```

Page B

```
MOD PAGE          CPU46%
   IN  OUT  RAT1
   72% 88% 0.35
DEP1 RAT2 DEP2  MCH
 18  0.62  9    4
```

- **cpu%** is the real per-block render load against the realtime budget. Under ~80 % is comfortable; sustained near 100 % means audio dropouts.
- **DECAY** shows estimated RT60 in seconds, or FRZ at full travel. Readings below about one second are rough estimates — trust your ears down there.

MIDI implementation

Port: TRS MIDI IN and OUT on the panel. **Channel:** set on Page B knob 7 (default 1), saved across power cycles. Only Control Change is used — no notes, no clock, no program change.

Control Change map

CC	Parameter	CC	Parameter
102	Decay	110	Input level
103	Damping	111	Output level
104	Lowpass	112	Mod rate 1
105	Highpass	113	Mod depth 1
106	Bitrate	114	Mod rate 2
107	Predelay	115	Mod depth 2
108	Diffusion	116	Bypass (≥ 64 = on)
109	Mix	65	Page (mirror only — see below)

All parameter CCs are 7-bit, 0–127, and use **the same mapping law as the physical knob**. CC 64 on the decay knob produces exactly what the knob at half-travel produces.

- **MIDI channel has no CC, deliberately.** Changing the receive channel over the channel you are currently listening on is a good way to lose contact with the module.
- **CC 65 (page) is logged but not obeyed.** The physical toggle is the only page authority; a software page change would be overridden on the next pass anyway.

How MIDI and knobs coexist

Last writer wins, per parameter. Each of the 14 parameters tracks its own owner independently:

- Send a CC → that parameter is MIDI-owned. The physical knob is ignored.
- Move the knob past a small deadband → the knob takes it back.
- Every other parameter is unaffected.

So you can automate reverb size from a DAW while riding MIX by hand.

Consequence worth knowing: *there is no pickup threshold on the takeback. A hand resting on a knob will fight DAW automation for that parameter. This was a deliberate choice for immediacy.*

MIDI output

Moving a knob transmits its CC. Three behaviours worth knowing:

- **Knob position is sent, not the mapped value.** A DAW recording and playing back a knob move reproduces the identical sound.
- **A knob resting at power-on transmits nothing.** The baseline is seeded on first movement, so powering up does not spray CCs.
- **MIDI-owned parameters do not transmit.** This is what prevents feedback loops when a DAW is both sending and recording.

Recording a patch to a DAW: sweep each knob once through its range with the DAW recording. Play it back and the module follows exactly. Because Page A and Page B use different CCs, both pages record independently regardless of which page is displayed.

Serial console

USB, 115200 baud. Quiet by design — nothing prints unless you ask or something happens.

Key	Action	Key	Action
1 !	Mod rate 1 down / up	sN	Save mod settings to slot N (0–3)
2 @	Mod rate 2 down / up	lN	Load slot N
q Q	Mod depth 1 down / up	nN...	Rename slot N
w W	Mod depth 2 down / up	B	List all four slots
c	CPU load + I ² C health	b	Toggle bypass
d	Print settings as pasteable source	h ?	Help

Preset slots

Four named slots store the four modulation parameters only — not gains, not MIDI channel, not any Page A setting. They are serial-only; there is no front-panel save.

Checking bus health

Press **c**:

```
cpu=41% (2400 us / 5805 us budget) blocks=123456
i2c: pot drops=0 / 1440000 reads (0 ppm) switch drops=0 [clean]
```

[clean] means every control read has succeeded since power-up. If you ever see **<-- BUS FAULT**, the panel controls are dropping reads — frozen knob values or spontaneous page changes are the symptoms.

Behaviour notes

- **Bypass priority.** With the hardware bypass switch fitted, the switch wins. Serial b and CC 116 still work, but a held switch re-asserts itself continuously — so if the toggle is down, remote bypass appears to do nothing.
- **What survives power-off:** the four mod parameters, MIDI channel, and the four preset slots. **What does not:** everything on Page A, plus input and output level. Page A follows its knobs at power-up, so it comes back exactly as the panel is set.
- **Freeze.** DECAY at full travel holds the tank indefinitely. Input still feeds in, so it layers rather than freezing a single snapshot.
- **Bitrate is backwards from what you would guess.** Fully clockwise is clean; turning it down adds crush. It is a fidelity control, not an effect amount.

Troubleshooting

Symptom	Check
No sound at all	Bypass toggle; MIX; IN LEVEL on Page B
Reverb but no dry signal	MIX fully wet
Distortion	IN LEVEL above 1.0 with a hot source; BITRATE not fully clockwise
Knobs do nothing	Wrong page — check the PAGE toggle and screen header
One knob does nothing	Page B knobs need moving past their stored value first
Knobs frozen	Press c; if drops are climbing, power-cycle
DAW automation fights you	Hand resting on that knob — last writer wins
Nothing on USB serial	115200 baud; console is silent by design, press h